Engineering Science

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B.Tech. in Engineering Science

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Message from Chairperson (ES Undergratuate Commitee)

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Engineering applications generally require multidisciplinary knowledge, spanning across various disciplines of engineering and sciences. Knowledge in humanities along with engineering and/ or sciences are also finding important applications. B.Tech. in Engineering Sciences (ES) is a unique bachelors degree program of IIT Jodhpur which aim to impart core knowledge in two targeted focus areas in Science and Engineering, to meet demands of emerging industrial and academic requirements.

Any student admitted to IIT Jodhpur can opt for B.Tech. in Engineering science post 1st year. Students will be required to complete core courses of ES, in the first two years of study whereas they can outline their own curriculum, in the guidance of faculty advisor, in the form of electives for the remaining two years, from courses across disciplines of engineering, science, and humanities. In line with regular B.Tech. programs, students can also use 10 credits out of stipulated electives and 10 additional credits to earn minor/

– Professor Ajay Agarwal (Chairperson)

Institute Background and Vision

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Innovations, Scientific Discoveries, and New Technologies are the essential foundations of the Indian Institute of Technology Jodhpur. The institute focuses on building solid and in-depth knowledge in the engineering domains and strongly encourages interdisciplinary learning. The courses offered are defined such that it enhances the imagination and creativity of the students to identify and solve new problems in society. Our endeavour is to build an atmosphere for uninhibited ideas.The institute offers 9 B.Tech programs: Computer Science & Engineering, Electrical Engineering, Engineering Sciences, Artificial Intelligence and Data Science, Mechanical Engineering, Bioengineering, Materials Engineering, Chemical Engineering and Civil and Infrastructure Engineering. The institute also empowers students by allowing them to avail specializations, minors, and elective courses.

Program Background

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Engineering Science is a unique branch of engineering designed to cater to the upcoming demand of the industry and research, training the best of students to have a strong foundation in every field along with a specialization in their topic of interest. Our institute is one of the primary institutes offering an Engineering Science program. This programme is a four-year course. The first two years are devoted to topics which we believe all Engineering undergraduates should study. In the third and fourth years, the students have the opportunity to study any two complementary disciplines by drafting their own course structure. This fosters multi-disciplinary thinking and integrated application of engineering, scientific and mathematical principles in young students. The students can specialise/minor in any field as well.

The final degree will read: B.Tech in Engineering Science with specialisation/minor

Engineering Science Options

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To provide flexibility in curricula, options have been created to pursue

curriculum in the form of electives for the remaining two years from courses across disciplines of engineering, science, and humanities. The students will be awarded B.Tech in Engineering Science. Similar to regular B.Tech. Programme Students can also use 10 credits out Of stipulated electives and 10 additional credits to earn minor/specialization as an add-on to the B.Tech degree.

- **4-year B.Tech. in X+Engineering Science :** Students can pursue a capability linked B.Tech. in X + Engineering Science to pursue newer Structures of programmes in addition to classical programmes, where 'X' represents the primary branch in which a student has taken admissions. After completion of the first year, the student having a CGPA of 6.5 or more may opt to choose Capability-linked B.Tech. in X+Engineering Science. Students will be required to register for extra credits for the same. The students pursuing this programme will take common courses and courses with a concentration in Engineering Science in the 2nd year. From the third year onwards the student would also start opting for courses related to the primary branch (X) and elective courses in the area of both primary branch and Engineering Science. At the end of the second year, an option of primary branch change will also be available to meritorious students of this programme. The students will be awarded B.Tech. in X and Engineering Science, e.g., B.Tech. in Mechanical Engineering And Engineering Science. Students of this programme would not be able to opt for any other specialization/minor.
- **3-year B.Sc. in Engineering Science:** This option is designed mainly for under performing students as an exit plan. The student would be required to complete required credits under institute and engineering science core, and open electives categories.

Domain Options for B.Tech. (ES)

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There are various combinations of complementary domains that a student can opt in their 5th and 6th semester. A few examples are: CS+EE, CS+BB, CS+Chem, CS+Math,CS+Phy, AI+BB, AI+EE.

CS+Chem

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Dancing Molecular: Study of properties of different elements for substituting Li-ion batteries using molecular dynamics. Molecular Dynamics is a computationally intensive task and required intense knowledge of Mathematics, Chemical Engineering, and Computer Science.

deals with the design of computer systems which involves the design of the new hardware or software where knowledge of electronics is prerequisite Also we can clearly understand about the networking and the telecommunication.

The objective of Engineering Science – Computer Science and Electrical Engineering program is to design the systems and components that capture, store, process, interpret, and transmit information or signals. Some of the most significant technological advances of the 20th century were either invented or put into practice by electrical engineers and computer scientists, including electric power systems; global broadcast and personal telecommunication systems; computer systems; computer networks; medical instrumentation, such as magnetic resonance imaging (MRI) and computer-aided tomography (CAT); integrated circuits; lasers; household appliances; and feedback control, such as for autopilots. New technologies developed by electrical engineers and computer scientists are likely to be even more important in the 21st century as a new era of intelligent, information-driven systems is made possible by fundamental advances in faster communication rates, smaller devices, and greater computational capabilities.

AI+EE

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The fifth generation (5G) wireless communications are expected to satisfy the diverse service requirements in various aspectsDue to the extreme range of 5G requirements,the design and optimization of 5G networks becomes very challenging. The future 5G network will require robust intelligent algorithms to adapt network protocols and resource management for different services in different scenarios. Artificial intelligence (AI), is a feasible solution for the emerging complex communication system design. It is now appropriate to apply AI technology to 5G wireless communications to tackle optimized physical layer design, complicated decision making, network management and resource optimization tasks in such networks

The objective of Engineering Science – Artificial intelligence and Electrical Engineering program is to design the systems and components that has provided a great potential and space for optimization in the field of electrical engineering. This can not only bring about a significant improvement not only with regards to the economic aspects but also in the safety and the actual control of the operation. Ever since artificial intelligence reached a stage where it could be used in various fields in order to make work simpler, it has been used widely in various fields of life, primarily in robotics and computer programming. Recent studies and investigations have shown that artificial intelligence can also be a viable option in the field of electrical engineering especially as a solution to some longstanding power system problems where conventional methods experience some difficulty.

each other. This process is especially prominent in systems such as those of DC or AC rotating electrical machines which can be designed and operated to generate power from a mechanical process (generator) or used to power a mechanical effect (motor).

Electromechanical Robotics is one of its applications.

Civil Engineering + AI

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With skylines of major cities around the world peppered with iconic buildings of all shapes and sizes, we can all agree that the boundaries and standards of design and engineering have been pushed beyond their limits. This is all thanks to the biggest game-changer in the industry—Artificial Intelligence in 3D Building Information Modeling (BIM).The construction process involves safety risks that can lead to actual accidents. AI provides the opportunity for more accurate data gathering from real-world context models to help civil engineers identify potential hazards in the construction process.Also, AI-enabled cameras and networks can continuously monitor all areas of construction and allow engineers to assess equipment use, track progress, and analyze activities in real-time, which helps in the early detection of potential risk hazards.

A good example of the application of such technology lies in the popularity of Indus.ai. This San Francisco tech company set AI-supported cameras around construction sites to provide real-time footage while collecting and evaluating data to give construction managers insights on things like the movement of materials and labor deployment at various areas of the site. This interactive dashboard also allows civil engineers to anticipate anything that may prove dangerous and make better decisions with regard to worker's safety.

Material Science + AI

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Materials science or solid-state physics is plagued by the 'curse of dimensionality'. This means even for a reasonable number of constituents of a material, the structure-property relationships are determined by a high-dimensional parameter space. These relationships are not something trivial to be elucidated only by experiments or theory. This is precisely where the applications of AI or machine learning (ML) come into play.

By leveraging machine learning, AI can manage disparate clinical trial datasets, enable virtual screening, and analyze vast amounts of data. Besides reducing clinical trial costs, AI can gain otherwise unobtainable insights and feed them back into the drug development process.

AI technologies to serve the biotech industry are being developed by several companies. Their services are rapidly becoming indispensable as older methods like classical statistical analysis or manual image scanning reach their practical limits.

Curriculum

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First and Second Semester

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Table-ES-Sem12 (use the horizontal scroll bar to see the full table)

1st Semester (ES)	Category	Course Name	L-T-P	Credits		2nd Semester (ES)	Category_	Course Name_	L-T-P_	Cred
1	IE	EEL1010: Introduction to Electrical Engineering	3-0-2	4		1	IE	MEL1010:Engineering Mechanics	2-1-0	3.0
2	IE	CSL1010: Introduction to Computer Science	3-0-2	4		2	IS	CYL1010:Chemistry	3-0-0	3.0

4	IS	MAL1010: Mathematics 1	3-1-0	4	4	IS	MAL1020: Mathematics II	3-1-0	4.0
5	IE	MEP1010: Engineering Visualization	0-0-2	1	5	IS	PHP1010:Physics Laboratory	0-0-2	1.0
6	NE	OAE1010: Engineering Design 1	0-0-2	1	6	IS	CYP1010:Chemistry Lab	0-0-2	1.0
7	NH	HSN1010: Communication Skill 1	0-0-2	1	7	IE	MEP1020:Engineering Realization	0-0-2	1.0
8	NH	OSN1010: Social Connect and Responsibilities	0-0-1	0.5	8	NE	OAE1020: Engineering Design II	0-0-2	1.0
9	NH	OSN1020: Performing Arts 1/ Sports 1	0-0-1	0.5	9	NH	HSN1020:Communication Skill II	0-0-2	1.0
			Total Graded	17	10	NH	OSN1040: Social Connect and Responsibilities	0-0-1	0.5
			Total Non-Graded	3	11	NH	OSN1050: Performing Arts 2/ Sports 2	0-0-1	0.5

					Total Non-Graded	3.0

Third and Fourth Semester

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Table-ES-Sem34 (use the horizontal scroll bar to see the full table)

3rd Semester (ES)	Category	Course Name	L-T-P	Credits		4th Semester (ES)	Category_	Course Name_	L-T-P_	Credits_
1	ES	CSL2020 Data Structuresand Algorithms	3-0-2	4		1	ES	CSL2050 Pattern Recognition and Machine Learning	3-0-2	4
2	ES	MAL2010 Probability,Statistics and Stochastic Processes	3-1-0	4		2	ES	CSL2060 Embedded Systems and IoT	3-0-2	4
3	ES	PHL2060 Modern Physics	3-0-0	3		3	ES	CSL2070 Modelling and Simulation	3-0-2	4
4	IE	EEL2010 Signals andSystems	3-1-0	4		4	ES	CSL2040 Design and	3-0-0	3

		Thermodynamics						Materials Selection		
6	IE	MTL2014 ComputationalMaterial Design	1-0-0	1		6	IE	MTL2012 Structure of Materials	1-0-0	1
7	NE	Intro. To Profession	0-0-2	1		7	IH	Humanities I	3-0-0	3
			Total Graded	20					Total Graded	20
			Total Non-Graded	1					Total Non-Graded	0

Engineering Science Undergraduate Committee

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It is a committee established to ensure the course is run efficiently and effectively.

- Prof. Ajay Agrawal,Dept. of Electrical Engineering,Chairperson
- Dr. Dip Shankar Banerjee,Asst. Professor, Dept. of CSE,Faculty Advisor
- Dr. Ankur Gupta,Asst. Professor, Dept. of Mechanical Engineering, Member
- Dr. Deepika Bhattu,Asst. Professor, Dept of Civil Engineering, Faculty Advisor
- Dr. Angan Sengupta,Asst. Professor, Dept. of Chemical Engineering, Time Table cordinator
- Dr. Ritu Gupta,Asst. Professor, Dept. of Chemistry, Member
- Dr. Prasenjeet A. Tribhuvan, Asst. Professor, Dept. of H&SS, Member

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- B20ES001 Adarsh Kumar __ CSE+EE (Communications)
- B20ES014 Akarshan __ CSE+EE (Communications)
- B20ES008 Samyak Bafna __ AI/DS + EE (Communications)
- B20ES004 Dhruv Agarwal __ AI/DS + Chemical
- B20ES005 Harsh Jaiswal __ CSE+EE (Communications)
- B20ES009 Utkarsh __ CSE+Chemical
- B20ES006 Narahari Sai Divya __ Bio Engineering + AI/DS
- B20ES007 Poonam Suthar __ BioEngineering + AI/DS
- B20ES016 Prakhar Katiyar __ CSE+Chemical
- B20ES013 Prashant Kumar __ CSE+EE (Communications)
- B20ES015 Sujal Gupta __ CSE+Chemical
- B20ES011 Vidhi Gupta __ CSE+Chemica
- B20ES010 Yash Anand __ CSE+Bio Engineering
- B20ES012 Yash Ruhatiya __ CSE+Bio Engineering

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